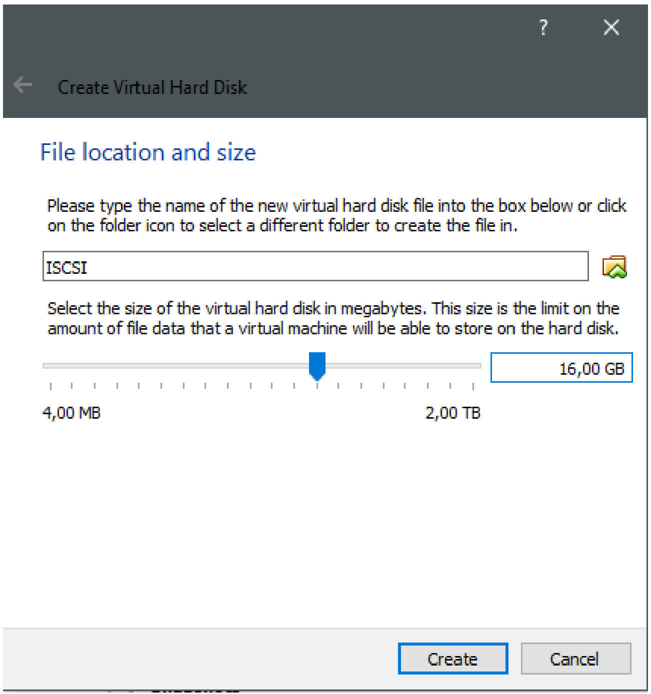
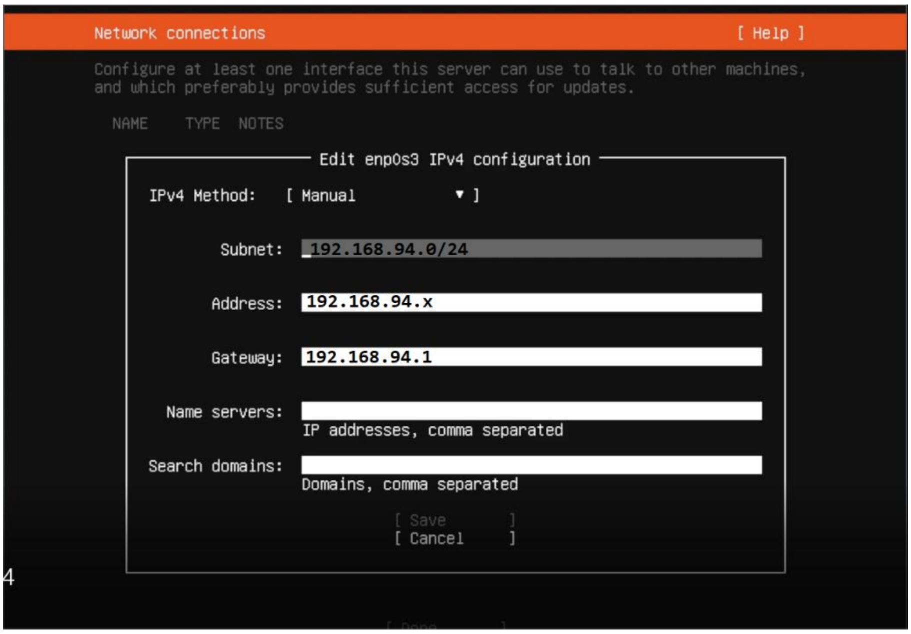
Laborator ISCSI
Ubuntu server 18.04

Începem prin a crea o mașina virtuala cu plăcile de rețea in modul Bridged si cu un Hard Disk de minimum 16 GB. Sistemul de operare este Ubuntul 18.04 server si va trebui instalat in mașina virtuala.



Atentie, la realizarea configurarii internetului se vor introduce urmatoarele:



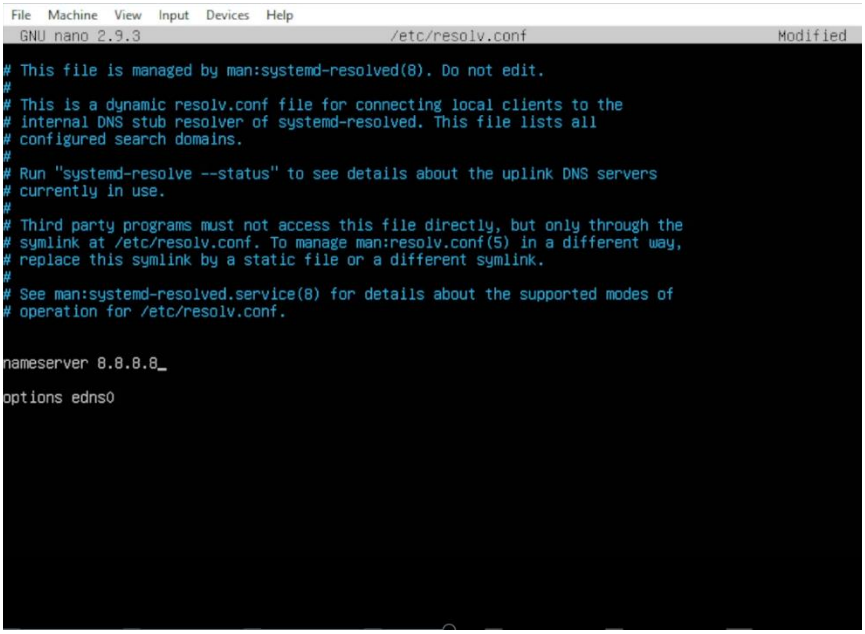
Unde x va fi ales in cadrul laboratorului astfel incat sa nu dea conflict cu celelalte adrese IP din laborator.

După instalare se va introduce următoarea comandă in modul administrator:

sudo apt-get update

In fișierul resolv.conf se afla informații despre serverele DNS necesare.

nano /etc/resolv.conf



apt -y install targetcli-fb

mkdir /tmp/iscsi_disks

Se porneste aplicatia pentru configurare ISCSI targetcli.

Atentie comenzile date in aceasta aplicatie sunt precedate de /> iar cele din line de comanda a sistemului de opereare de #! In aplicatia targetcli functioneaza comenzile uzuale de cd si ls pentru a naviga in structura de directoare.

targetcli

```
root@cata:~# targetcli
Warning: Could not load preferences file /root/.targetcli/prefs.bin.
targetcli shell version 2.1.fb43
Copyright 2011-2013 by Datera, Inc and others.
For help on commands, type 'help'.

/> ls
o- / ......................................................................... [...]
  o- backstores .............................................................. [...]
  | o- block .................................................. [Storage Objects: 0]
  | o- fileio ................................................. [Storage Objects: 0]
  | o- pscsi .................................................. [Storage Objects: 0]
  | o- ramdisk ................................................ [Storage Objects: 0]
  o- iscsi ............................................................ [Targets: 0]
  o- loopback ......................................................... [Targets: 0]
  o- vhost ............................................................ [Targets: 0]
/> _
```

Exista mai multe tipuri de stocare pentru ISCSI:

An administrator can choose any of the following backstore devices that Linux-IO (LIO) supports:

`fileio` backstore

Create a `fileio` storage object if you are using regular files on the local file system as disk images. For creating a `fileio` backstore, see Creating a fileio storage object.

`block` backstore

Create a `block` storage object if you are using any local block device and logical device. For creating a `block` backstore, see Creating a block storage object.

`pscsi` backstore

Create a `pscsi` storage object if your storage object supports direct pass-through of SCSI commands. For creating a `pscsi` backstore, see Creating a pscsi storage object.

`ramdisk` backstore

Create a `ramdisk` storage object if you want to create a temporary RAM backed device. For creating a `ramdisk` backstore, see Creating a Memory Copy RAM disk storage object.

In acest laborator vom folosi tipul fileio, adica un fisiere pe HDD:

backstores/fileio create iscsi_file /tmp/iscsi_disks/disk01.img 1G

```
/> backstores/fileio create iscsi_file /tmp/iscsi_disks/disk01.img 1G
Created fileio iscsi_file with size 1073741824
```

iscsi/

```
/> iscsi/
```

create iqn.2018-08.com.example.storagesrv01.targe01

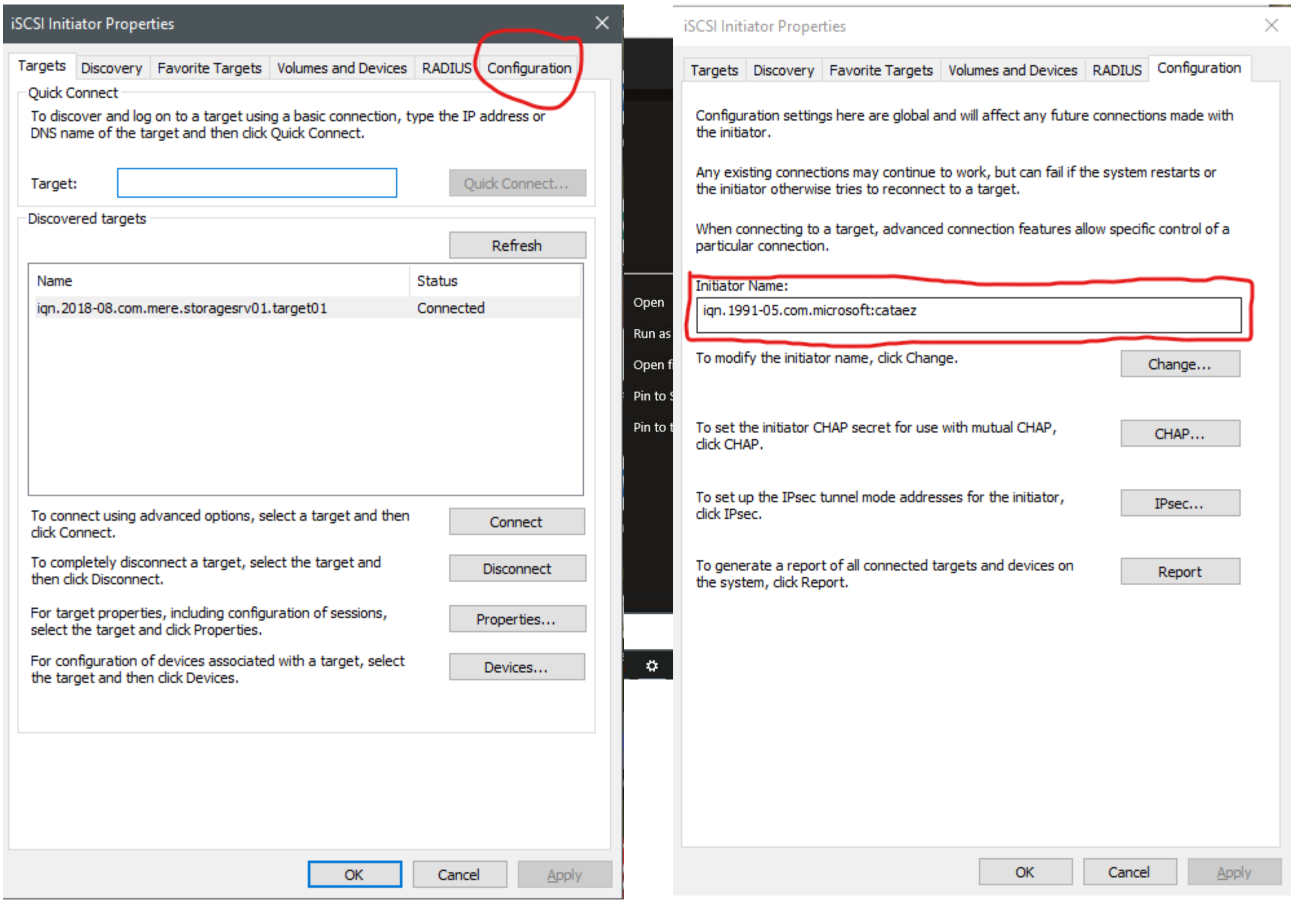
iqn.2018-08.com.example.storagesrv01.targe01/tpg1/

luns/ create /backstores/fileio/iscsi_file

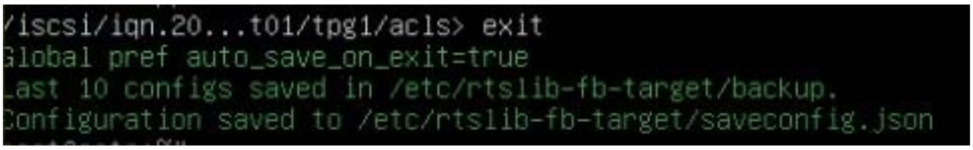
acls/

Create „initiator name”

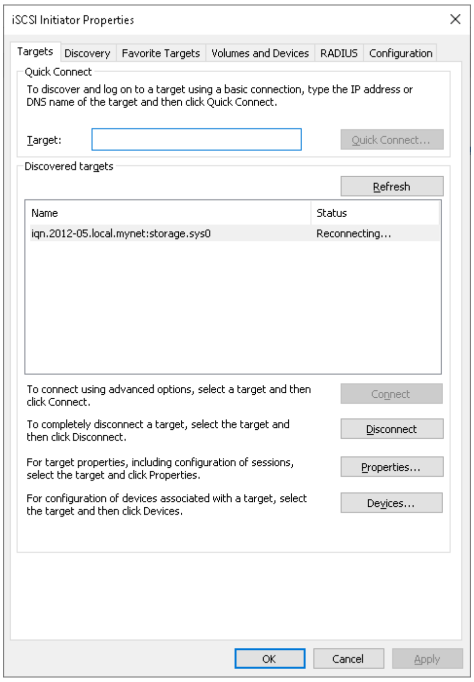
Acest initator name se gaseste astfel



exit



La final se realizează connectarea cu ajutorului iSCSI Initiator-ului din Windows la IP-ul masinii virtuale anterior creata.

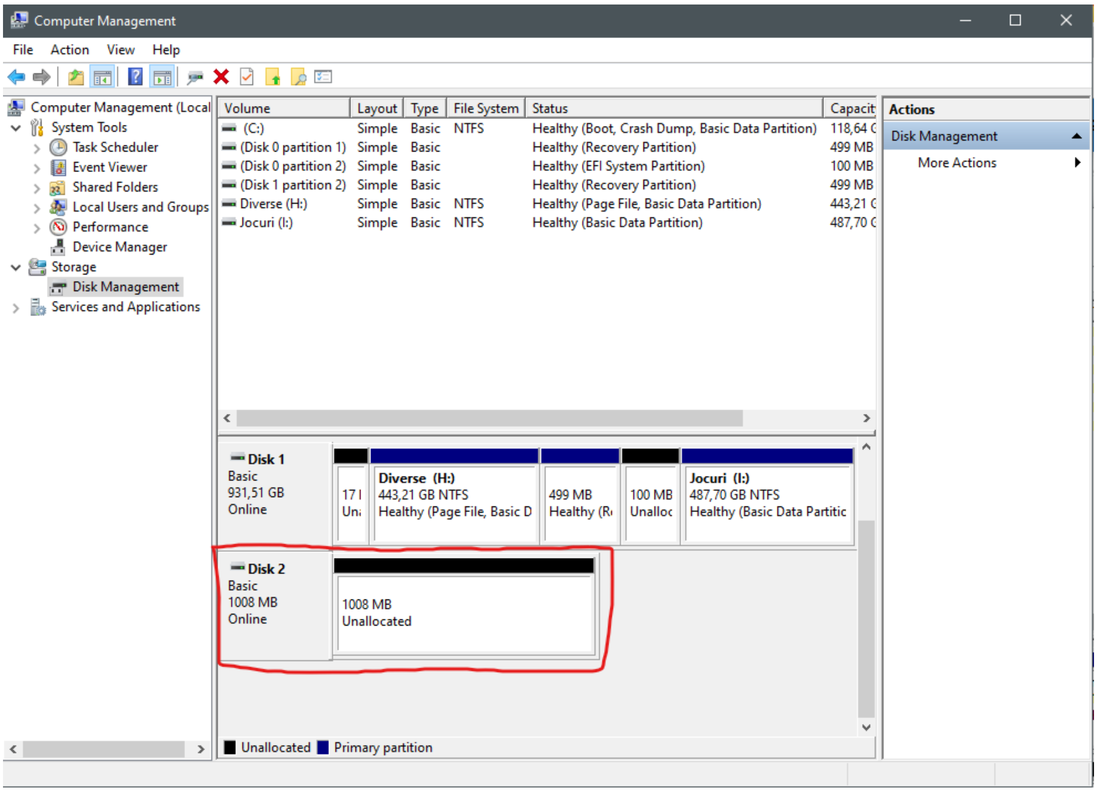


Pentru a nu interveni probleme, este recomandata dezactivarea Firewall-ului cand incercati connectarea sau permiterea portului `3260` .

Se apasa butonul *Connect* si i**n cazul in care conectarea a avut succes apare la status** ***Connected*** **,**

In continuare, pentru a configura o partitie pe noul disc conectat prin ISCSI, trebuie accesata fereastra Computer Management (Click dreapta This PC > Manage > Storage)

Aici se poate observa o partite de exact 1GB nealocata.



Anexa:

Alte detalii despre procesul de instalare le găsiți aici:

https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/9-beta/html/managing_storage_devices/configuring-an-iscsi-target_managing-storage-devices